



CastaliaOS^{98 PE}

Powerful Edition · an original retro desktop OS

An original, Win9x/XP-inspired retro desktop operating system for DOS-class PCs — written from scratch in C. No Microsoft code, art, or branding.

By **Dave Abellan & Claudio di Castello**

Studio **Tombatossals Softworks**

Price **Free** · MIT

Status **Public preview** · v0.1.0

Platforms **DOS 386+** · Emulators

Written in **C** · ~30.9k lines

The tour

A quick animated look at the desktop, the apps, and the live benchmark filling in — all rendered by the OS's own software renderer.



CastaliaOS 98 PE
Powerful Edition · an original retro desktop OS

an original retro desktop OS · runs on a 386
by Dave Abellan & Claudio di Castello

Animated tour (GIF) — desktop, File Manager, Media Player, Solitaire, the Benchmark Suite, About, and more. (animated in the online kit)

About CastaliaOS

Remember beveled buttons, a Start menu, glossy title bars, and a file explorer with a friendly blue task pane? CastaliaOS recreates that feeling as a **brand-new**, **legally-clean** system you can actually run on period hardware.

CastaliaOS 98 PE ("Powerful Edition") is a from-scratch desktop operating system that captures the look and feel of late-90s / early-2000s computing while being completely original. Hand-written in portable C, it boots on a 386, composites its entire UI with an original software renderer, and ships a real desktop shell: a Start-style launcher, a Windows XP-style file explorer, four virtual desktops, window snapping, and a themeable toolkit — with 20+ apps on top, including a Winamp-style WAV player, a benchmark suite, and Klondike Solitaire.

It is free and open source under the MIT license, and it develops and unit-tests on a modern "host" build so the whole stack stays verifiable — while shipping to real DOS via Open Watcom.

~30.9k

lines of C

~116

source files

457

tests passing

20+

applications

6

architecture layers

4

virtual desktops

386+

target hardware

MIT

open source

Fact sheet

Title	CastaliaOS 98 PE ("Powerful Edition")
Creators	Dave Abellan & Claudio di Castello
Studio	Tombatossals Softworks (independent)
Type	Original retro desktop OS / hobby operating system
Release	In development — public preview (v0.1.0, MVP / Phase 1)
Platforms	Real DOS (386+), DOSBox, 86Box, PCem, QEMU; headless host build for CI
Price	Free — open source (MIT)
Language	English UI, source & docs
Website	tombatossalssoftworks.com
Contact	hello@tombatossalssoftworks.com

What makes it notable

- 100% **original code and artwork** — a real, runnable retro OS, not a theme or a mod.
- Runs on **genuine DOS hardware**, yet develops and tests on a modern host build.
- An **original software renderer** with XP-style glossy chrome, drop shadows, and dirty-rect compositing in a tiny memory budget.
- **Extensible**: discoverable `.CAPP` add-ons appear as first-class launcher apps — something Windows 98 SE never had.
- **Four virtual desktops**, window snapping/tiling with a live preview outline, right-click menus, and live theming.

Features

Desktop & shell

- Two-column XP-style Start panel with full keyboard navigation
- Taskbar with Quick Launch, system tray, and a clock + date tooltip
- Four instant virtual desktops; window snapping & tiling; draggable desktop icons
- Right-click context menus; live-recoloring themes (incl. XP-style **Aurora**)
- Six bundled wallpapers with a live picker & monitor preview
- Software cursor, drop shadows, open/close/minimize animations, screensaver

Window manager & toolkit

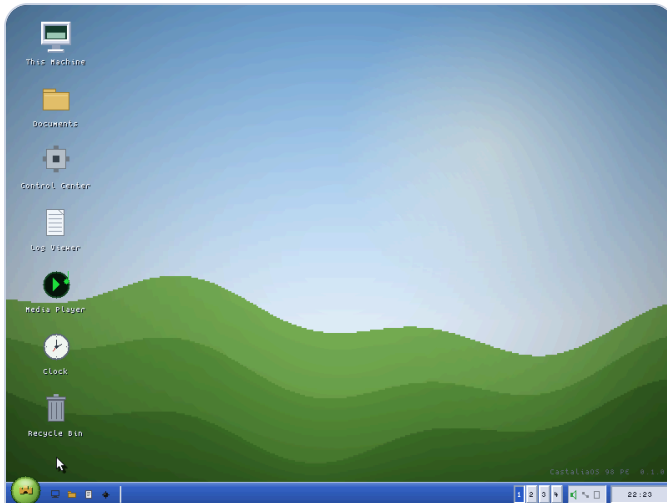
- Real WM: frames, caption buttons, focus/z-order, Alt+Tab, resize
- Original UI toolkit: buttons, checks, radios, lists, menus, text fields, dialogs
- Shared system clipboard across apps

Applications

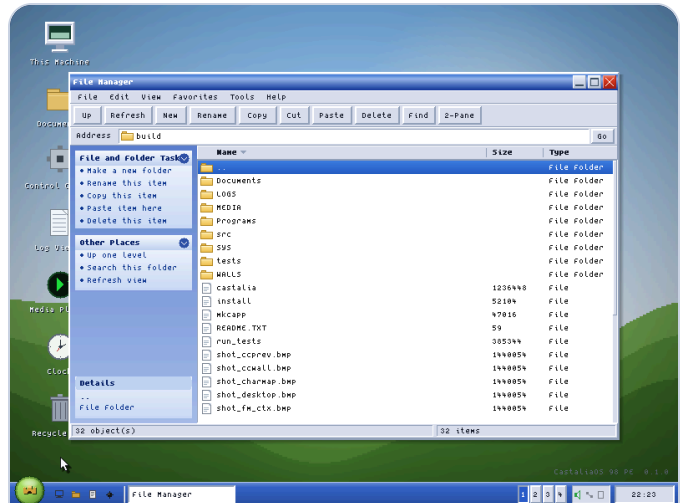
- **File Manager** — XP-style Explorer: address bar, blue task pane, 3 views, sortable columns, context menus, Properties, search, two-pane mode
 - **Media Player** — Winamp-style WAV player with a live visualizer, a drag-to-reorder playlist, shuffle/repeat, folder scanning
 - **CastaliaWrite** — a word processor: menus, flat toolbars, ruler, page view, word-wrap, bold/italic/underline, alignment, save/load
 - **CastaliaSheet** — a three-sheet workbook with a real formula engine: references, ranges, SUM/AVG/MIN/MAX/COUNT/ABS/INT/ROUND, Auto-Sum, CSV save/load, error cells
 - **Benchmark Suite** — CPU/memory/graphics tests with a CastaliaMark score
 - **About** — tabbed hardware & live source-code statistics
 - **Solitaire** (Klondike), **Clock & Calendar**, **Character Map**, **Mines**
 - Notepad, Calculator, Task Manager, Paint-lite, Viewer, Log Viewer, Control Center, System Information
-

Screenshots

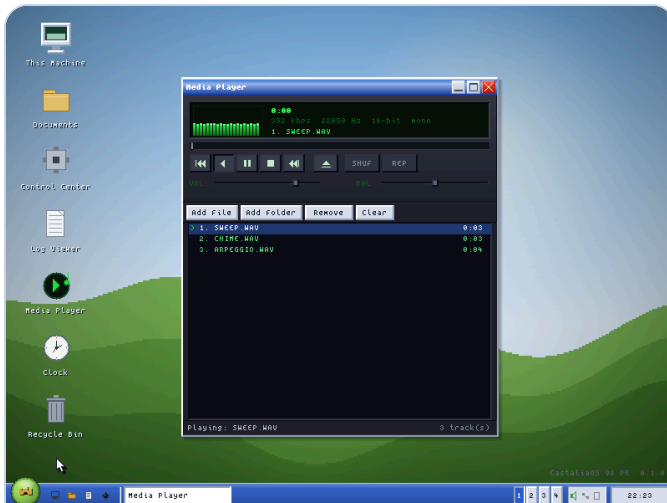
Native 800×600 captures. Full-resolution PNGs are in the `screenshots/` folder. 100% original UI and artwork.



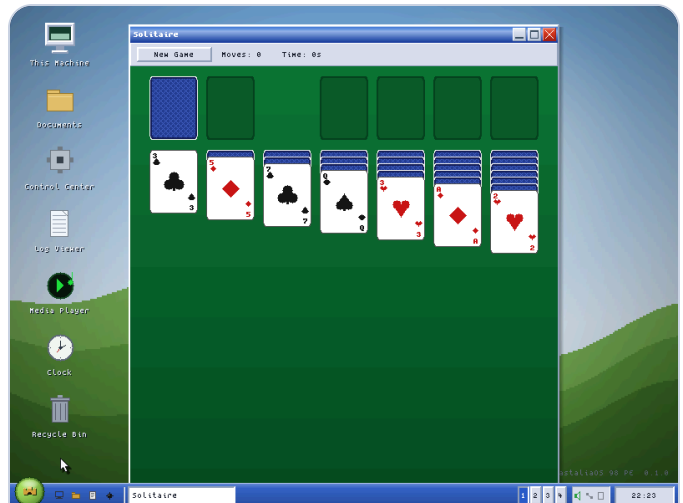
The desktop — Aurora theme, "Castalia Hills" wallpaper.



File Manager — XP-style Explorer with a blue task pane.



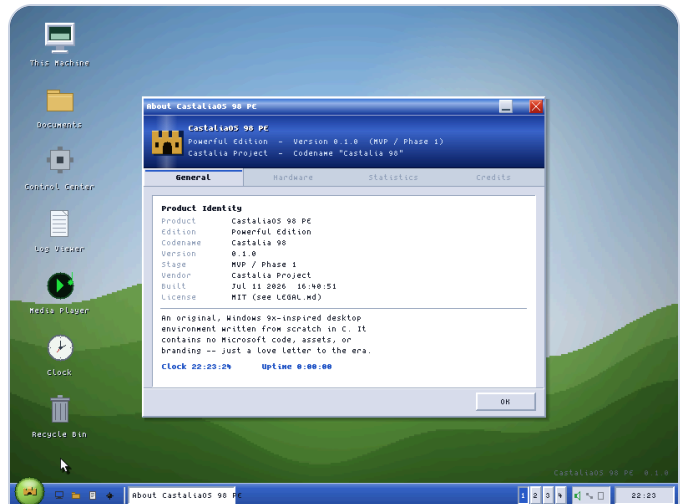
Media Player — a live WAV visualizer and playlist.



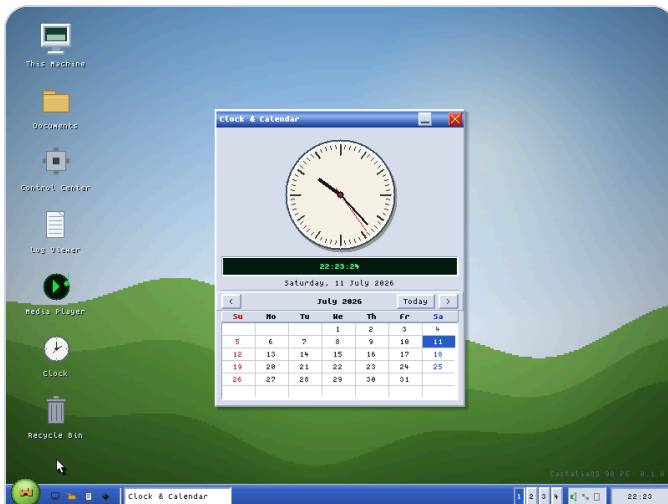
Solitaire — Klondike with drag-and-drop and a bouncing-cards win.



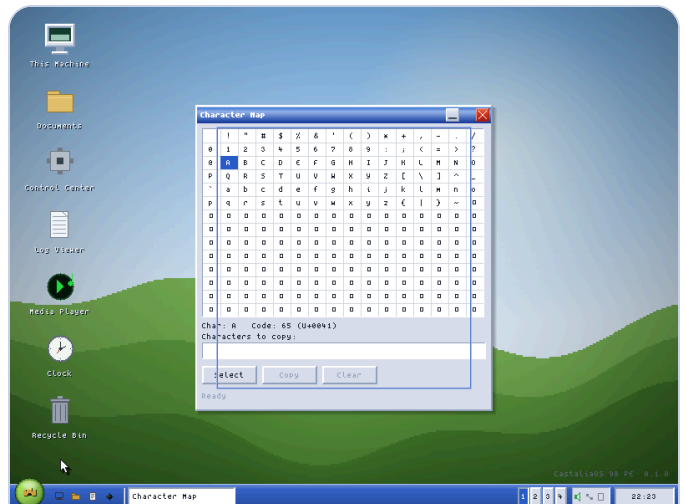
Benchmark Suite — CPU/memory/graphics + CastaliaMark.



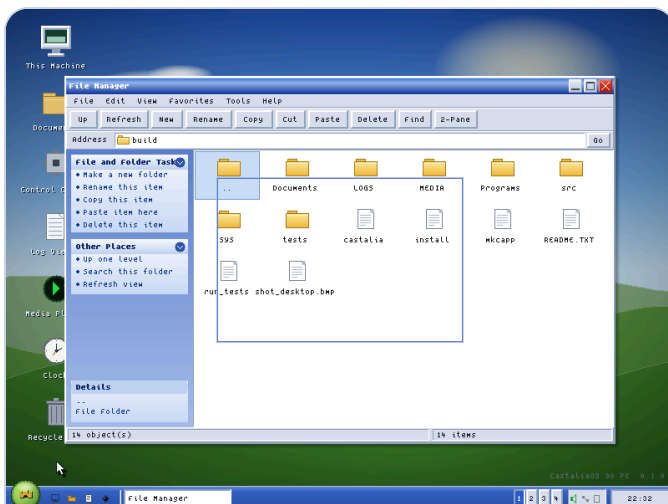
About — hardware and live source-code statistics.



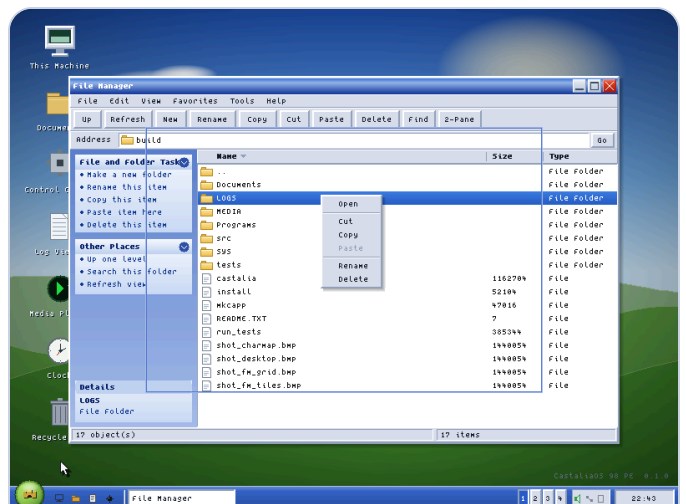
Clock & Calendar — a live analog clock.



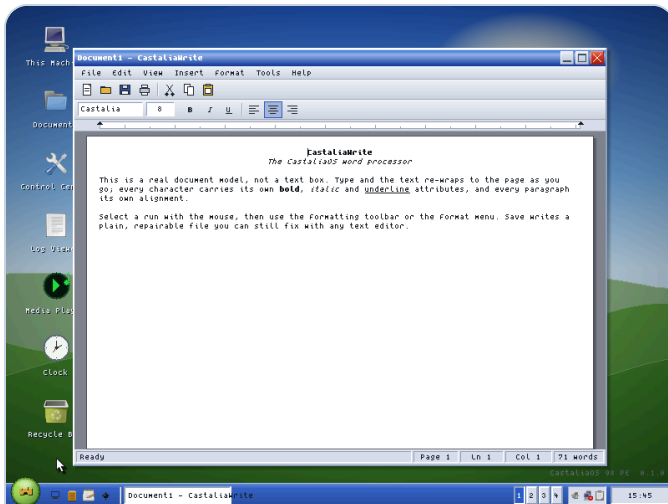
Character Map — pick glyphs and copy them.



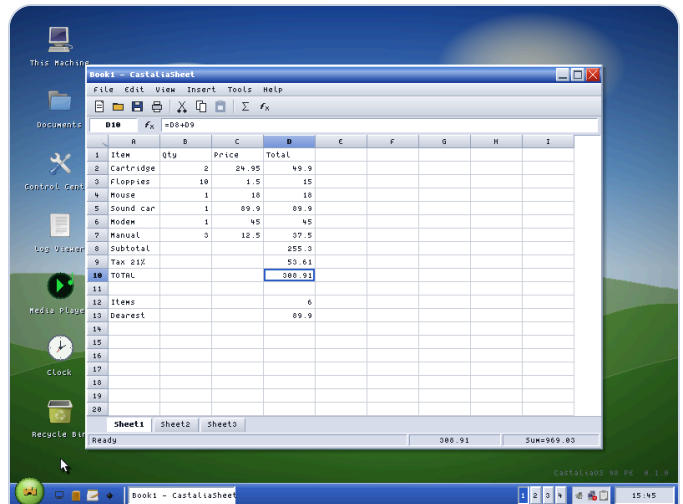
File Manager — the Icons (grid) view.



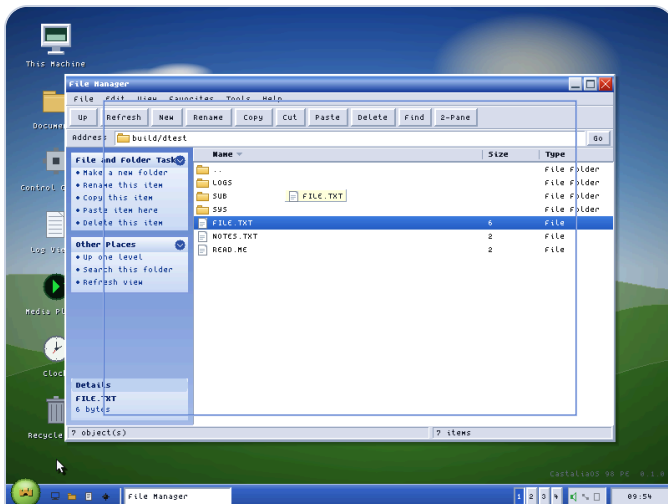
Right-click context menus, XP-style.



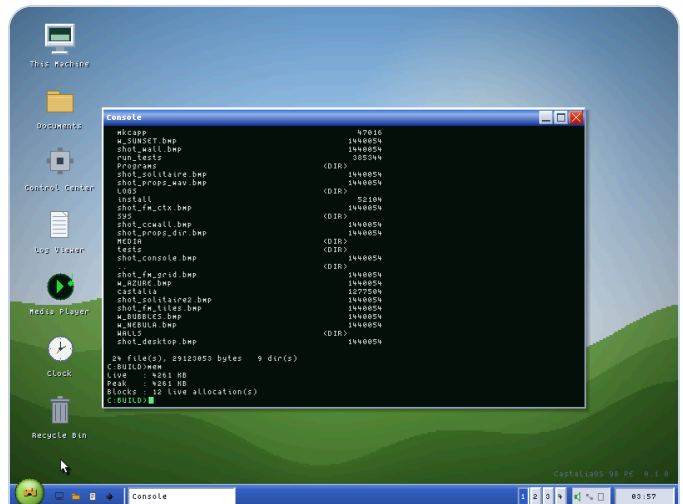
CastaliaWrite — menus, toolbars, ruler and page view.



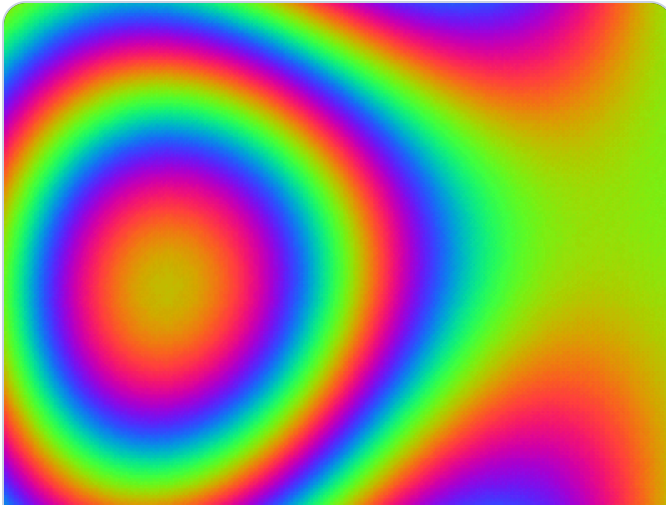
CastaliaSheet — Name Box, fx, sheet tabs and a live column sum.



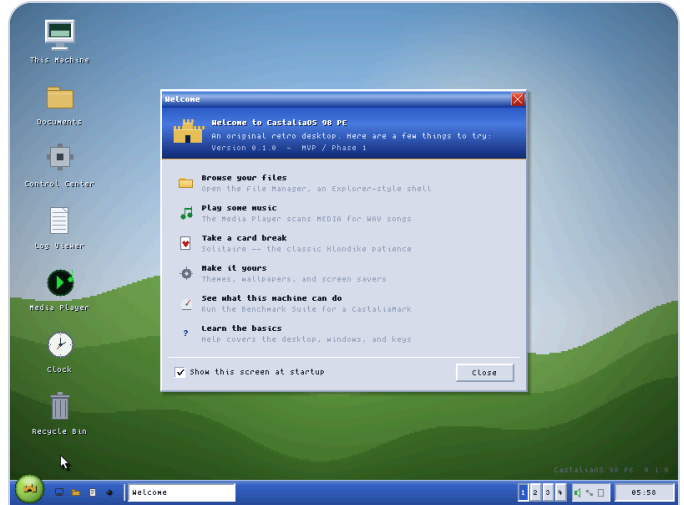
Drag-and-drop — a floating ghost follows the cursor.



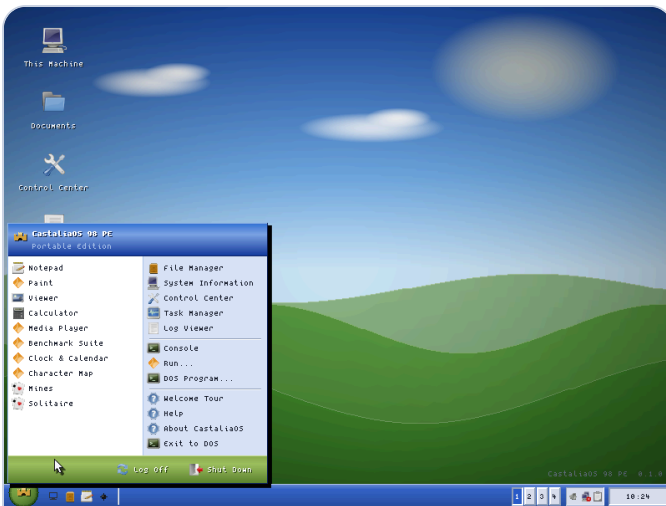
Console — a DOS-style command shell with real built-ins.



Screensavers — a demoscene plasma (also starfield & Mystify).



The Welcome tour — first-boot links to the key apps.



The Start-style launcher menu.

Technical

A small, layered design in portable C. Nothing above the platform layer touches the hardware — which is what lets the same stack build on DOS and on a host.

The stack we used

- **C (strict C89)** for the portable core; Watcom-compatible
- **Open Watcom** for the DOS target (DOS/4GW, 32-bit protected mode)
- Any C89 compiler for the **headless host** build (CI + screenshots)
- Small Python helpers author fonts/icons offline
- Zero runtime dependencies; bundled **Spleen** font (BSD), optional **Tango** icons

The rendering model

The whole UI composites into one **32-bit XRGB back buffer** in system RAM. Only `present()` converts the dirty rectangles to real hardware — a search-free **3-3-2 palette** at 8bpp or **RGB565** at 16bpp. No per-pixel alpha: a 1-bit magenta color key for icons/cursors and cheap edge-strip shadows keep XP-style chrome inside a period memory budget.

Verified by 457 host unit-test checks, a strict-C89 (Watcom) lint over every portable source, and a golden-screenshot pipeline.

Six layers: Platform → System → Graphics → Window Manager → UI → Shell → Apps. Extensible via a documented `.CAPP` add-on ABI.

Roadmap

Now — v0.1.0

- Bootable, themeable desktop
- WM, UI toolkit, virtual desktops, snapping
- 20+ apps, incl. XP file manager & media player
- 457 tests + golden screenshots

Next — Phase 2–3

- Drag-and-drop in the file manager
- Real Sound Blaster DAC audio streaming
- ARP/IP/UDP/TCP over the packet-driver seam
- Richer Paint, a terminal, a welcome tour
- Theme editor & downloadable packs

Later — Phase 4+

- Friendly installer + bootable CD/USB image
- A catalog for `.CAPP` add-ons
- A small third-party SDK
- Localization & low-end tuning

The dream: an original desktop a community can extend — where a `.CAPP` you wrote on a modern laptop drops onto a 386 and just runs, and the whole thing still fits on a couple of floppies.

Press release & assets

Everything here is cleared for editorial use. Please credit **Tombatossals Softworks** and link the project.

This kit as one file: [Download the press kit \(PDF\)](#)

Documents: [Fact sheet](#) [Descriptions](#) [Features](#) [Technical](#) [Roadmap](#) [History](#) [Press release](#) [Contact](#)

Logos: [Logo \(SVG\)](#) [Logo \(PNG\)](#) [Logo — dark bg \(PNG\)](#) [Crest icon 256](#) [Crest icon 64](#)

Get in touch

Review builds, interviews, and extra assets on request.

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